VIT BHOPAL

"Cafe Management System - Running Technology"

Cafe Management System

Fundamentals of AI and ML

Registration no.:
25BCE11035

Faculty:
M.K.Jayanthi

TRUSHA SHIVAJI DHABE

Introduction

This project is focused on the creation of a Smart Cafe Management System (CAFEM) using the Python programming language. The intention of this project is to enhance various aspects of daily café operation through improved management functions, enhanced feature sets, intelligent feedback suggestions, and well-organized development practices. By demonstrating the ability of using software application to create a more efficient operational process for small businesses, the project aims to present the benefits of improved electronic processing to enhance small café operations through the application of electronic technology in a simple console-based application environment.

Problem Statement

When you run a cafe, you have many responsibilities such as managing your menu, taking customer orders, tracking sales, and analyzing customer behavior. Some of these responsibilities can be done manually or without structure. This often causes problems like confusion, slow service, and errors. This project will address these problems with a structured and automated solution that will improve the efficiency of your business while enhancing the overall experience for customers.

Functional Requirements

The system should be able to:

- Add, update, view, and delete menu items
- Allow users to place and manage orders
- Generate bills/receipts for customers
- Suggest items based on popularity or past behavior
- Classify customers into categories (VIP, Regular, etc.)
- Simulate and display sales data

Non-Functional Requirements

- The system should be easy to use and understand
- It should handle invalid inputs without crashing
- Code should be clean, modular, and maintainable
- Performance should be efficient for small-scale usage
- The system should be reliable for repeated operations

System Architecture

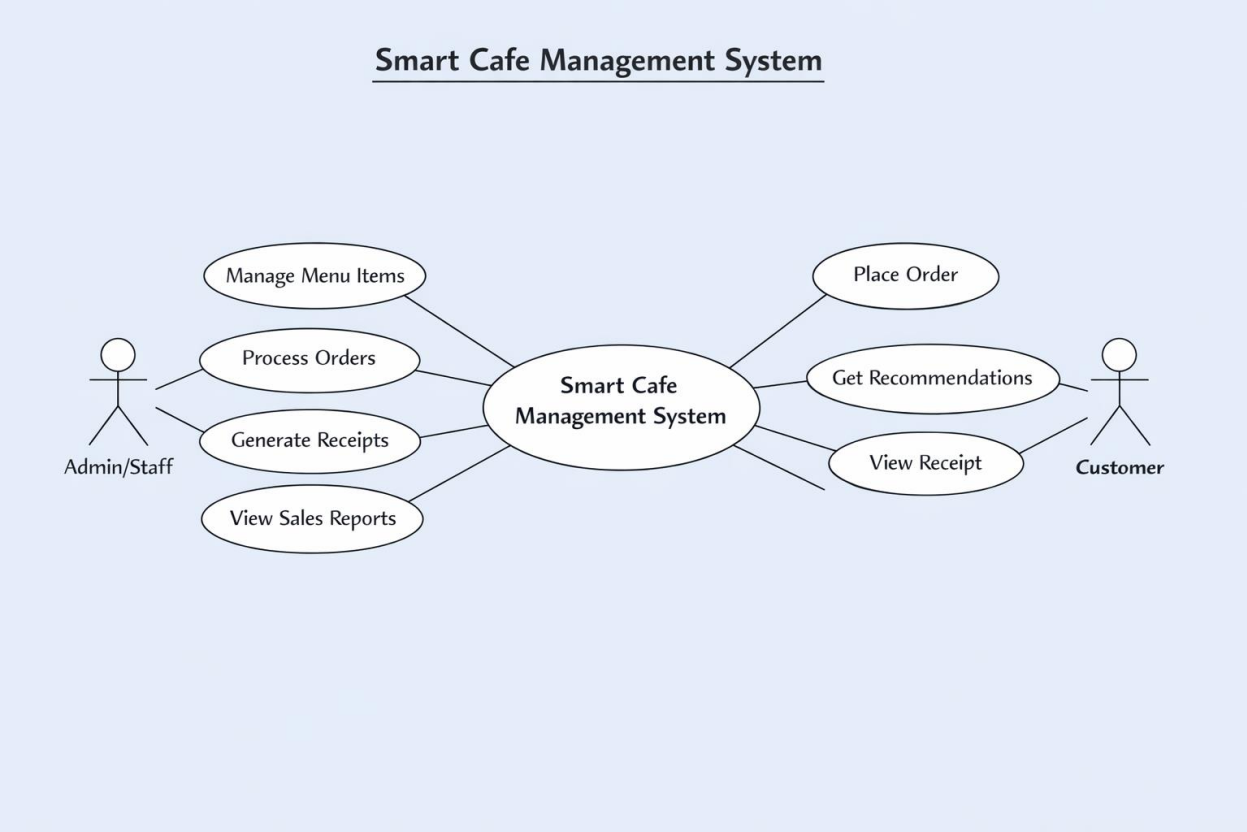
The system follows a modular structure where different parts of the program handle specific responsibilities.

- Data Layer: Manages storage and retrieval of menu and order data
- Business Logic Layer: Handles operations like recommendations, billing, and classification
- Presentation Layer: Interacts with the user through the console

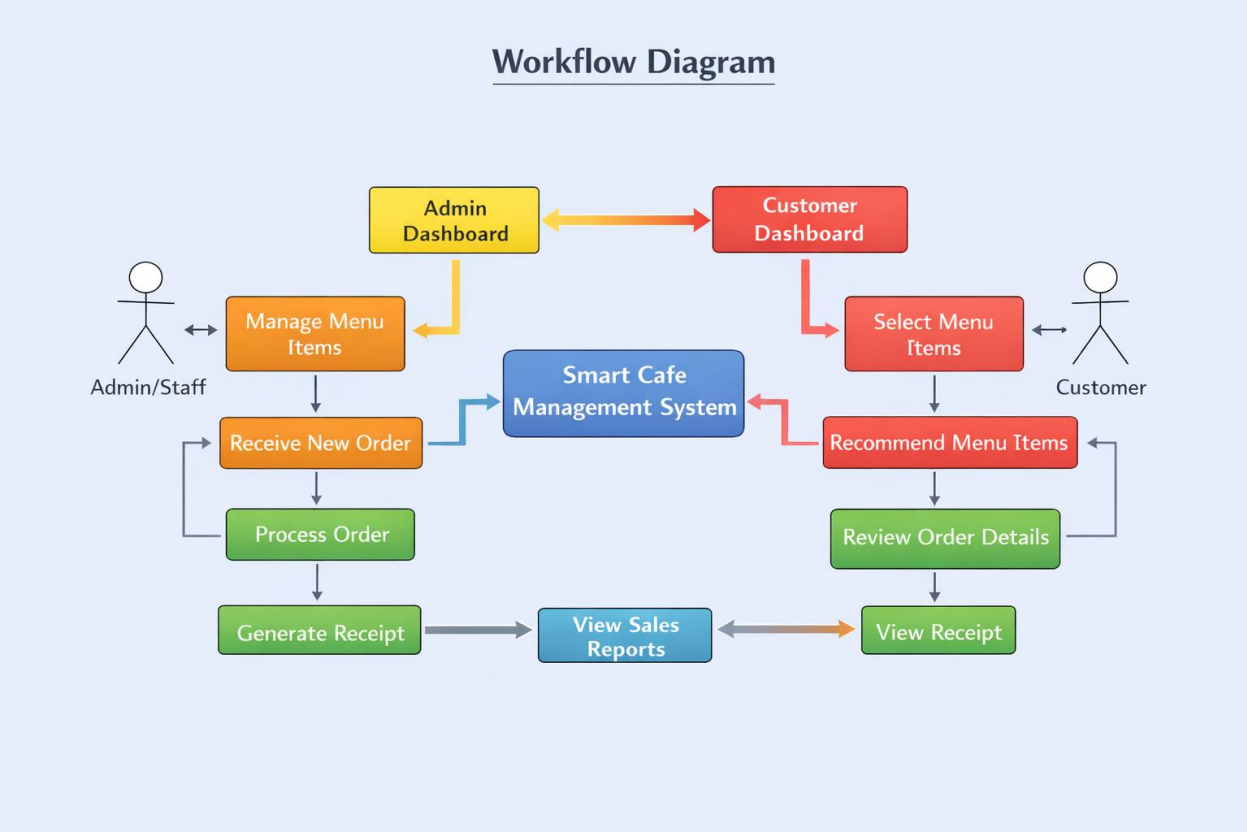
This separation makes the system organized and easier to extend.

Design Diagrams

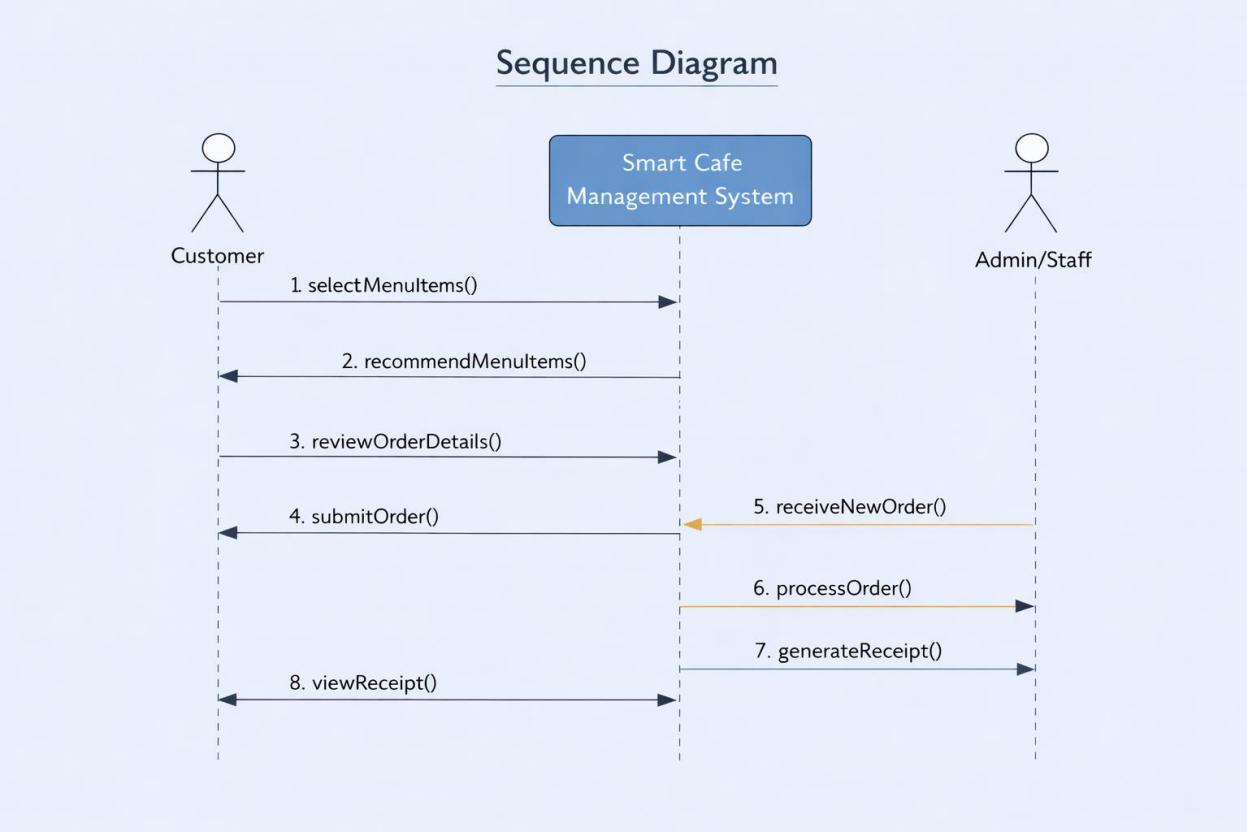
Use Case Diagram



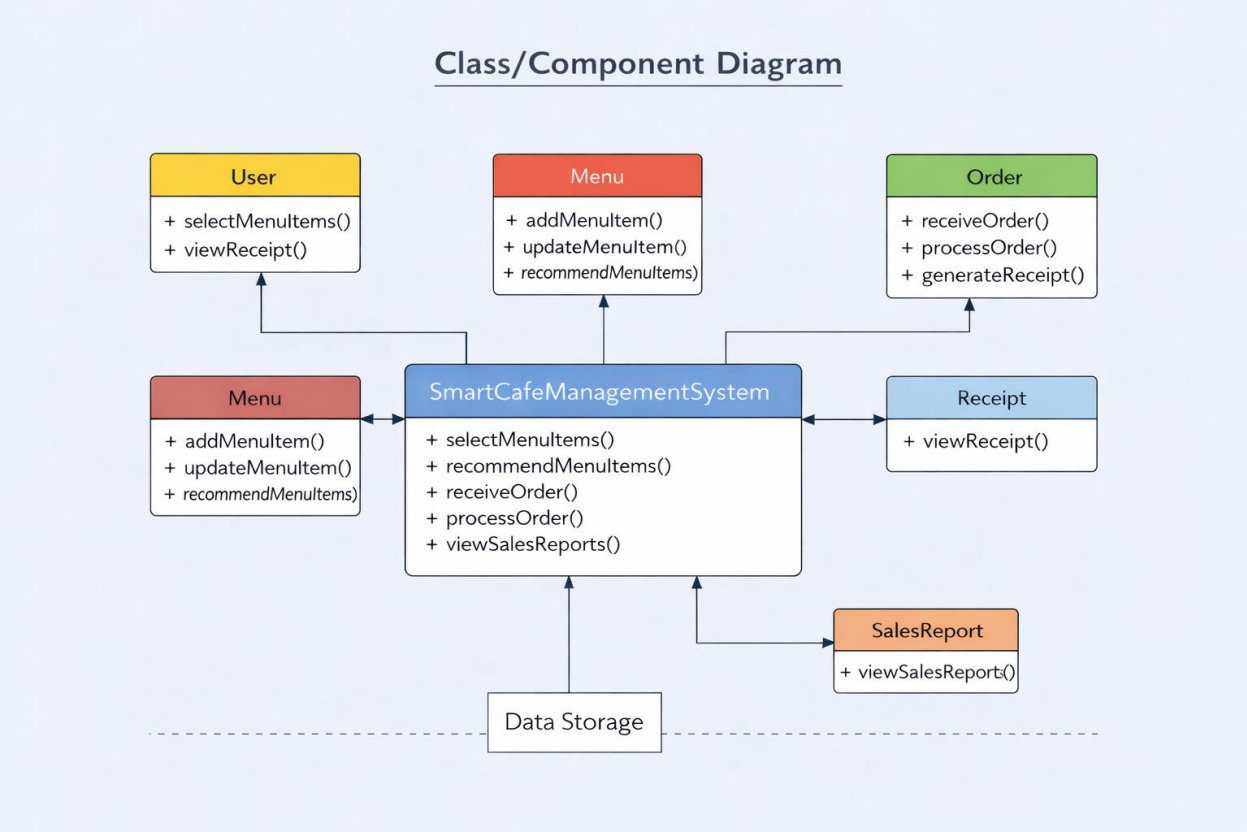
Workflow Diagram



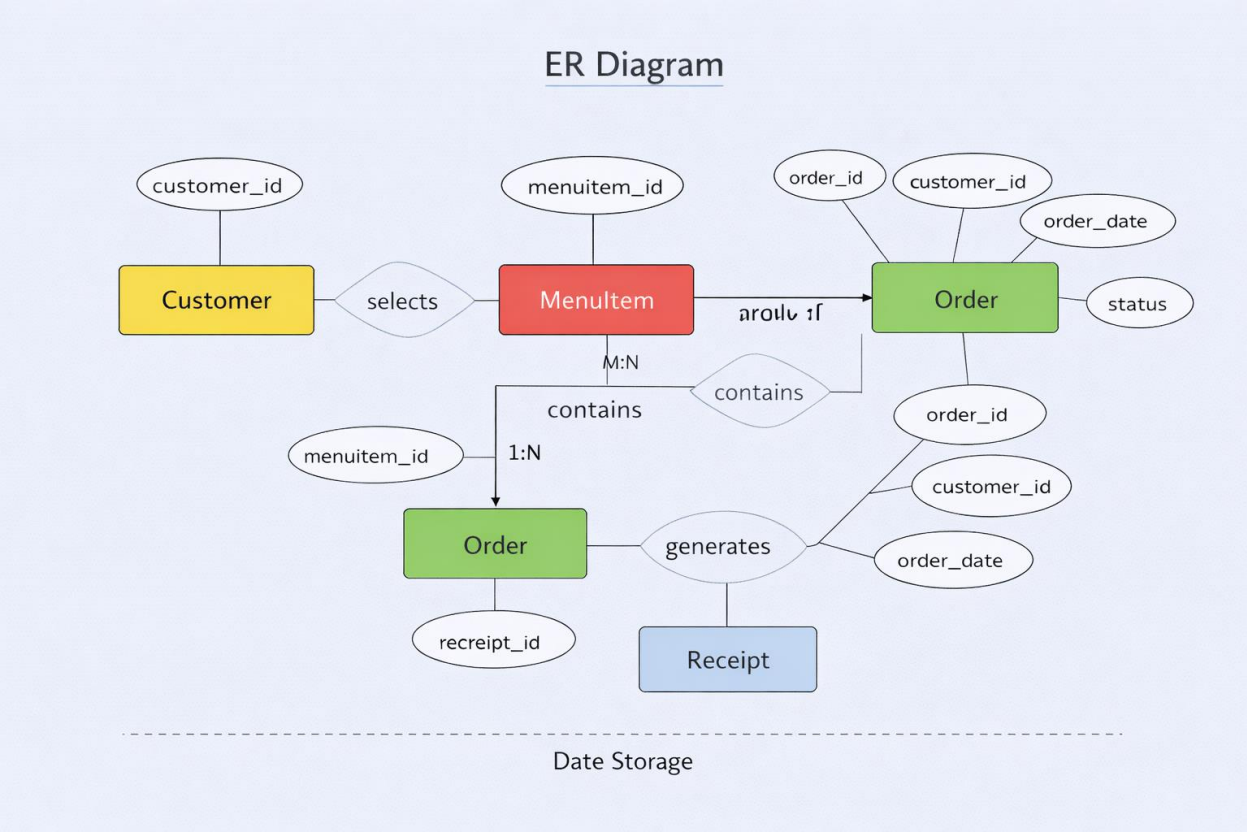
Sequence Diagram



Class/Component Diagram


Class/Component Diagram
User
+ selectMenuItems()
+ viewReceipt()
Menu
+ addMenuItem()
+ updateMenuItem()
+ recommendMenuItems()
Order
+ receiveOrder()
+ processOrder()
+ generateReceipt()
Menu
+ addMenuItem()
+ updateMenuItem()
+ recommendMenuItems()
SmartCafeManagementSystem
+ selectMenuItems()
+ recommendMenuItems()
+ receiveOrder()
+ processOrder()
+ viewSalesReports()
Receipt
+ viewReceipt()
SalesReport
+ viewSalesReports()
Data Storage

ER Diagram



Design Decisions & Rationale

A modular design separates different functionality from one another so they can each be managed separately. The use of design patterns such as Repository Pattern to organize how the code accesses data helps increase the maintainability of the application and allows it to evolve over time, while the Strategy Pattern facilitates changes to the recommendation logic.

Implementation Details

This project is developed in Python, paying attention to structure as well as how easy it will be for people who read this codebase to understand what's happening. There are a variety of tools built into the application (e.g., error handling, logging, type hints). Additionally, the application is broken down into several modules to help clarify the codebase structure and maintain separation of responsibilities.

Screenshots / Results

```
/usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"
macbook@Macbooks-MacBook-Air untitled folder % /usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"
==================================================
        SMART CAFE MANAGEMENT SYSTEM
==================================================

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 1
Customer name: Soumya

--- MENU ---
1. Coffee     - Rs 50
2. Tea        - Rs 30
3. Pastry     - Rs 200
4. Finish Order
5. Get Recommendation
Please select (1-5): 3
Enter quantity for Pastry: 2
Added 2 x Pastry - Rs 400

--- MENU ---
1. Coffee     - Rs 50
2. Tea        - Rs 30
3. Pastry     - Rs 200
4. Finish Order
5. Get Recommendation
Please select (1-5): 4

==================================================
                CAFE BILL RECEIPT
==================================================
Customer: Soumya
Date: 2026-03-26 21:22:20
--------------------------------------------------
Item         Qty  Price    Total
--------------------------------------------------
Pastry        2    Rs200    Rs400
--------------------------------------------------
GRAND TOTAL                Rs400
==================================================

Customer Status: Occasional Customer
2026-03-26 21:22:20,647 - INFO - Order saved: Soumya

Based on your preferences, try our Pastry!
Thank you for your order!

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 2

--- BUSINESS ANALYTICS ---
Total Orders: 1
Total Revenue: Rs 400
Unique Customers: 1
Average Order Value: Rs 400.00
```

```
macbook@Macbooks-MacBook-Air untitled folder % /usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 2

--- BUSINESS ANALYTICS ---
Total Orders: 1
Total Revenue: Rs 400
Unique Customers: 1
Average Order Value: Rs 400.00

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 3

--- 7-DAY SALES SIMULATION ---
Day 1: Rs 3193
Day 2: Rs 4943
Day 3: Rs 2196
Day 4: Rs 4446
Day 5: Rs 3517
Day 6: Rs 6275
Day 7: Rs 4197

Analytics:
   Average: Rs 4110
   Best: Rs 6275
   Worst: Rs 2196
   Total: Rs 28767

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 4
Enter customer name: Trusha

Recommendation for Trusha:
Popular choice: Pastry!

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 5
```

```
macbook@Macbooks-MacBook-Air untitled folder % /usr/local/bin/python3 "/Users/macbook/untitled folder/KBC.py/sourcecode.py"
Day 1: Rs 3193
Day 2: Rs 4943
Day 3: Rs 2196
Day 4: Rs 4446
Day 5: Rs 3517
Day 6: Rs 6275
Day 7: Rs 4197

Analytics:
   Average: Rs 4110
   Best: Rs 6275
   Worst: Rs 2196
   Total: Rs 28767

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 4
Enter customer name: Trusha

Recommendation for Trusha:
Popular choice: Pastry!

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 5

--- ADMIN MENU ---
1. Add Item
2. View All Items
3. Update Item
4. Delete Item
5. Back to Main Menu
Choice: 3
Item ID to update: 4
Error: Item not found!

--- ADMIN MENU ---
1. Add Item
2. View All Items
3. Update Item
4. Delete Item
5. Back to Main Menu
Choice: 5

MAIN MENU
1. Take New Order
2. View Analytics
3. Sales Simulation
4. Get Recommendations
5. Admin Menu
6. Exit System
Enter your choice (1-6): 6

==================================================
            CAFE SYSTEM CLOSED
==================================================
Final Report: 1 orders, Rs 400 revenue
Thank you!
macbook@Macbooks-MacBook-Air untitled folder %
```

Testing Approach

The system is tested using different input scenarios to ensure it behaves correctly. This includes:

- Valid and invalid user inputs
- Edge cases like empty orders or unavailable items
- Repeated operations to check stability

Challenges Faced

- Designing a clean and modular structure
- Handling unexpected user inputs properly
- Implementing recommendation logic in a simple way
- Keeping the system realistic while staying within scope

Learnings & Key Takeaways

- Improved understanding of real-world application design
- Learned how to structure code using design patterns
- Gained experience in handling user input and errors

- Understood how software can solve practical business problems

Future Enhancements

- Add a graphical user interface (GUI)
- Integrate a real database for persistent storage
- Include online ordering and payment features
- Improve recommendation system using advanced algorithms

References

- Python official documentation
- Online tutorials and learning resources
- Basic concepts of software design patterns